

TCPFMC: Trustworthy Cyclic Progressive Fusion for Multimodal Classification

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Abstract—The exponential growth of multimodal data has facilitated significant advances in multimodal classification techniques. Most multimodal classification algorithms aggregate information from different modalities into a unified representation to capture the holistic information. However, existing fusion mechanisms overly rely on high-quality data, which compromises the robustness of the model. Meanwhile, the fusion process can cause the loss of modality-specific information due to inherent modality differences. In this work, we propose a trustworthy cyclic progressive fusion method for multimodal classification, named TCPFMC. To enhance robustness and reduce reliance on high-quality data, we designed a modality energy score to evaluate the confidence of each modality, quantifying the informativeness of the modalities. In addition, for efficient integration of modality-specific information, we propose a novel cyclic progressive fusion approach that combines modality information at a fine-grained level, enhancing the overall performance of the model. Extensive experiments on six multimodal datasets show that TCPFMC outperforms state-of-the-art methods. The original code is available at <https://github.com/aoli-hrbust/TCPFMC>

Index Terms—Confidence evaluation, multimodal classification, progressive fusion, trustworthy learning.

I. INTRODUCTION

WITH the increasing availability of multimodal data, multimodal learning has achieved remarkable success [1], [2], [3], [4]. In real-world scenarios, multimodal learning is important in various fields, such as sentiment analysis

[5], [6], medical diagnosis [7], [8], and image classification [9], [10]. By utilizing the complementarity of modalities [11], models can capture more comprehensive information to improve performance (e.g., combining image and text data can provide richer semantic insights than a single modality alone [12]). However, existing multimodal methods lack an effective confidence evaluation mechanism and suffer from information loss during the fusion stage, which limits their practical application in fields such as medical diagnostics.

In multimodal learning, previous multimodal fusion methods have mainly focused on feature-level fusion and decision-level fusion [13], [14]. Feature-level fusion uses information aggregation to map multiple modalities to the same feature space for integration. Several typical methods use cross-modal label fusion to capture the relationships between different modalities and labels [15], [16]. Similar works aim to identify common information across different data types as the final representation [17], [18]. Compared with feature-level fusion, decision-level fusion is more computationally efficient, as it avoids deep joint modeling of raw data by making separate predictions for each modality and then aggregating them into the final decision. Several representative methods adopt decision-level fusion to integrate multisource evidence while quantifying predictive uncertainty, typically relying on dynamic feature and modality selection to drive fusion and final decision-making [19], [20]. Because fusion is performed only after each modality produces an independent decision, the model cannot capture deep feature interactions or fully leverage cross-modal correlations and diversity. In addition, while these methods outperform unimodal approaches, they tend to focus too heavily on the shared information from dominant modalities, overlooking the modality-specific information. Therefore, efficiently extracting the useful information from each modality is crucial. Trustworthy multimodal learning is a viable solution to address the above issue. It uses different trust strategies to calculate the uncertainty of information in various parts of the model, enhancing the interpretability of the overall computation process. In recent years, various methods have emerged to calculate modality-level reliability or overall trustworthiness. Some studies utilize true class probability (TCP) to dynamically evaluate both feature-level and modality-level information, thereby enhancing the model's ability to aggregate information [21], [22]. Another approach measures confidence by evaluating attention scores from both local and global perspectives, allowing the model to focus

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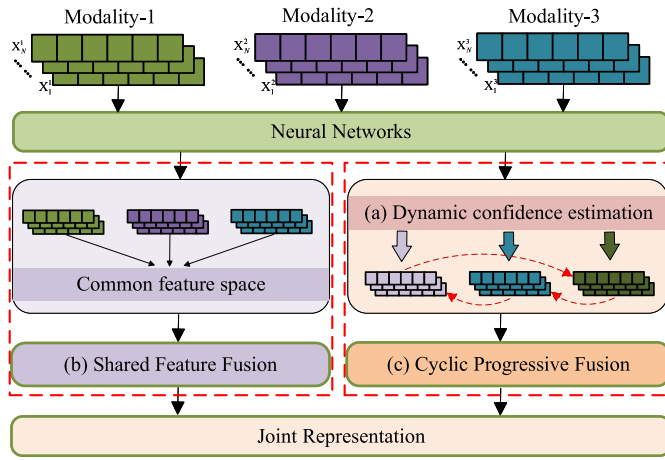


Fig. 1. (a) Illustration of modality confidence evaluation. The usefulness of information varies across modalities and samples. (b) Most methods for modality fusion primarily focus on mapping modality features into a shared feature space. (c) To reduce the loss of modality-specific information, our method proposes a cyclic progressive fusion mechanism.

more precisely on important features [23]. In addition, a multilevel confidence learning method has been introduced, combining feature confidence and label confidence to perform trustworthy information aggregation [16].

Despite the fact that trustworthy multimodal methods reduce model uncertainty, most approaches evaluate modality credibility based on the classification probability distribution. These methods still face several challenges as follows.

- 1) *Poor Noise Robustness*: The quality of multimodal data varies and contains a large amount of noise, making it difficult to effectively assess the confidence of each modality.
- 2) *Modality-Specific Information Loss*: As shown in Fig. 1(b), during the fusion stage, the process of extracting common features often fails to capture modality-specific information, leading to the loss of modality-specific information.

To address the aforementioned issues, we propose trustworthy cyclic progressive fusion for multimodal classification (TCPFMC), which performs progressive and modality-specific fusion guided by dynamically estimated modality-level confidence. Compared with MMDynamics [22], we develop a modality energy score to quantify the information content of each modality without relying on true class probabilities. However, to enhance robustness in confidence estimation, we introduce a modality-specific memory network that facilitates reliable confidence modeling under noisy or redundant inputs. Finally, to mitigate the loss of modality-specific information during fusion, we propose a cyclic progressive fusion scheme that enables gradual pairwise interactions among modalities. In this way, TCPFMC preserves shared information while effectively injecting discriminative, modality-specific cues into the fused representation. Extensive experiments demonstrate the effectiveness of both the confidence mechanism and the fusion strategy. Our method achieves optimal performance on all six benchmark datasets.

In summary, the contributions of this work are as follows.

- 1) Design a novel trustworthy multimodal fusion framework that estimates the information content of each modality and effectively integrates the modality-specific confidence information.
- 2) Propose a concise modality energy score that dynamically evaluates the confidence information of each modality based on its modality-specific patterns.
- 3) Propose a cyclic progressive fusion mechanism that integrates modality-specific information while preserving shared information.
- 4) Extensive experiments demonstrate that TCPFMC outperforms other state-of-the-art algorithms on benchmark datasets.

In Sections I–IV, we first review related work on multimodal learning and trustworthy learning, then provide a detailed description of the proposed method, and finally present extensive experimental analysis to validate the effectiveness of the approach.

II. RELATED WORKS

In this section, we review related work on multimodal learning and trustworthy learning, with a focus on multimodal fusion methods and trust evaluation mechanisms. Concurrently, we present representative state-of-the-art multimodal fusion methods from recent years in Table I.

A. Multimodal Learning

Compared to unimodal learning, multimodal learning effectively leverages the complementary information from different modalities to represent the overall features of samples [1].

Early multimodal fusion methods typically concatenate on data level [24], [25], [26], [27]. Intermediate fusion methods primarily aggregate modality feature information at the representation layer, achieving fusion by reconstructing overall information within the latent space [28], [29], [30], [31], [32]. Qu et al. [33] achieve efficient fusion and intelligent diagnosis through a quantum convolutional neural network (NN) and a variational quantum classifier. Chang et al. [34] proposed a dual-branch Transformer framework that effectively captures hierarchical and complementary information in multimodal remote sensing classification. Hu et al. [35] proposed a multimodal recommendation framework using GNNs with modality-independent receptive fields and a global transformer to better capture information across modalities. Decision-level fusion methods [15], [36], [37] typically make separate predictions for each modality and use the confidence level of each modality to calculate the uncertainty for modal fusion. Lu et al. [20] enhanced the interpretability and reliability of the model results by introducing dynamic feature selection to fuse decision information from different modalities. Zheng et al. [16] proposed feature confidence learning to enhance discriminative features and used label decision loss to promote separable representations. Multilevel fusion methods combine the advantages of feature-level and decision-level fusion, which can better adapt to the characteristics of different modes and have stronger robustness [38], [39].

TABLE I
COMPARISON OF FUSION PARADIGMS IN MULTIMODAL LEARNING METHODS

Methods	Early fusion	Intermediate fusion	Decision fusion	Dynamical fusion	Progressive fusion
SVM [25], KNN [26], RF [27], LR, NN	✓	✗	✗	✗	✗
GMU [31], Unit [39], GRidge [29] BPLSDA [30], BSPLSDA [30]	✗	✓	✗	✗	✗
Self-MM [37], MSCAE [36], MOGONET [15]	✗	✗	✓	✗	✗
TMC [46], ETMC [19], QMF [47]	✗	✗	✓	✓	✗
MoTIF [20], MLCLNet [16]	✗	✗	✓	✗	✗
MMDynamics [22], CEM [48] DPNET [50], DMIB [29], ACMVL [51]	✗	✓	✗	✓	✗
TCPFMC	✗	✓	✗	✓	✓

In recent years, dynamic fusion methods [40], [41], [42] facilitate adaptive information integration by considering the real-time information status or evolving decision trends of various modalities. Wang et al. [43] proposed a variational dynamic fusion network that utilizes multiscale spatial graphs and Gaussian mixture constraints to learn adaptive fusion patterns. Yi et al. [44] proposed a parallel deep learning model with dynamic weighting to fuse multimodal molecular features. Zhu et al. [45] significantly improved accuracy and robustness by integrating multipath-like data augmentation and a dynamic fusion mechanism. However, existing dynamic fusion methods often overlook modality-specific cues and lack mechanisms for progressive and hierarchical integration, which limits their ability to capture deep cross-modal dependencies.

B. Multimodal Trustworthy Learning

In multimodal environments, variations in modality quality can result in the neglect of critical information from certain modalities [51]. This makes the incorporation of a reliability mechanism particularly essential. In recent years, trustworthy multimodal learning has been proposed in various forms.

Bayesian-based methods quantify uncertainty with distributions in place of fixed parameters [52], [53]. Such methods allow for a more flexible representation of uncertainty and enable better generalization in uncertain or noisy environments. Ensemble-based methods [54], [55] utilize multiple deterministic NN models to estimate the uncertainty in the prediction results, reducing overfitting by aggregating diverse model predictions. Gal and Ghahramani [56] proposed to effectively model the prediction uncertainty of deep learning models by using Gaussian processes and dropout. Han et al. [46] model class probabilities and uncertainty with variational Dirichlet distributions and fuse modalities at the evidence level to improve robustness. Huang et al. [57] propose a deep evidential fusion framework that uses Dempster–Shafer theory with contextual discounting to quantify uncertainty and improve accuracy. Zou et al. [49] achieved reliable classification in biomedical applications by modality confidence estimation, and label confidence estimation. Zhu et al. [50] proposed an adaptive learning method with confidence fusion to evaluate confidence metrics and perform weighted integration. Unlike uncertainty assessment

algorithms, confidence calibration methods [21], [58] primarily calibrate classification results to obtain confidence. Han et al. [22] use TCP [21] to obtain modal confidence and dynamic fusion of modal features at the representation level. Li et al. [48] use normalized cross-entropy loss to assess confidence, while Kastanos et al. [59] use a bi-directional approach for lattices for confidence evaluation. Zhang et al. [47] propose a theory-driven framework that uses decision-layer energy estimation for dynamic, precise multimodal fusion. Yan et al. [60] propose a credible dynamic feature fusion network with uncertainty-aware fusion for incomplete multimodal breast ultrasound data.

In summary, existing confidence estimation methods are often task-dependent. They overlook the intrinsic quality of modality features, especially in cases where modality noise or semantic misalignment arises at the representation level prior to classification. This motivates us to estimate modality reliability from the embedding space and perform fusion in a progressive and hierarchical manner.

III. METHOD

This section details the proposed method. As shown in Fig. 2, TCPFMC follows a three-step pipeline: 1) the modality energy score module evaluates the reliability of each modality from its latent representation, providing confidence weights; 2) these weighted features are fed into the cyclic progressive fusion module, which performs pairwise adversarial alignment to progressively integrate multimodal information; and 3) the fused representation, together with outputs from modality-specific classifiers, is used by a fusion classifier for the final prediction. The detailed process is shown in Algorithm 1.

We define the multimodal data as $\mathcal{X} = \{X_1, X_2, \dots, X_M\}$, where M represents the number of modalities. For the m th modality, we have $X_m \in \mathcal{R}^{N \times d_m}$, where N represents the number of samples, and d_m denotes the feature dimension. The n th sample of the m th modality is represented as x_n^m , and the corresponding class label is denoted as $y_n \in \{1, \dots, K\}$, where K represents the number of classes. In the multimodal classification process, the encoder network is defined as $E_m : X_m \rightarrow Z_m$, where E_m maps the input modality data X_m to the encoded representation Z_m . At the same time, a separate classifier is designed for each modality, denoted as

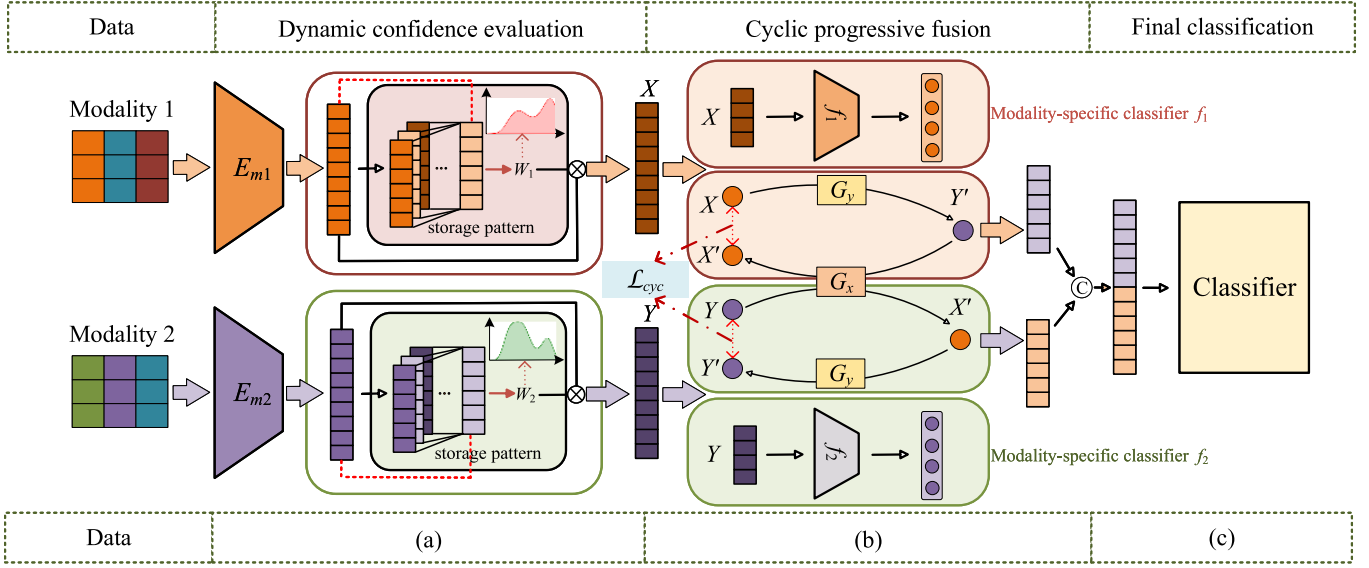


Fig. 2. Trustworthy multimodal fusion framework. The methodology consists of the following three main steps. (a) Modality energy score estimates the information content of each modality. The potential representation of the modal data is obtained by E_m , and the energy score of the current modality is calculated as W_m . (b) Under the action of the modality-specific classifier, each modality's potential representation performs cyclic progressive fusion. (c) Fused features are generated as the final joint representation for the final multimodal classification task.

$f_m : x_n^m \rightarrow y_n$, which maps x_n^m to its corresponding class label, y_n .

We first introduce a modality energy score to evaluate information content (see Section III-A for details). Then, modality-specific classifiers stabilize latent representations. Finally, a cyclic progressive fusion approach preserves key information while progressively integrating modalities (see Section III-B for details).

A. Modality Confidence Evaluation

For trustworthy multimodal classification methods, determining the confidence of each modality is crucial [22], [61]. Existing confidence assessment methods struggle to provide reliable modality information, resulting in significant redundancy and noise during fusion. To address this, we design a modality energy score to evaluate the information content of each modality. Our memory-based confidence estimation is conceptually inspired by Hopfield networks [62], [63], but differs in both objective and implementation. Unlike Hopfield networks that focus on global state retrieval, our method emphasizes local distributional alignment between input samples and modality centers.

1) *Hopfield Network*: The Hopfield Network is a classical feedback NN that can store and retrieve continuous patterns. By minimizing a predefined energy function, the input pattern $\xi_m \in \mathbb{R}^{d_m}$ can be gradually updated to a converged pattern that is similar to one of the stored patterns. The latent representation Z_m obtained from the encoder network E_m is used as the input pattern ξ_m . However, we define the stored patterns for each modality as $S \in \mathbb{R}^{d_m \times K}$, where each column $s_j \in \mathbb{R}^{d_m}$ represents the stored pattern corresponding to a particular class for this modality. The energy function is defined as

$$\text{Energy} = -\text{LSE}(\beta, \xi_m^T S) + \frac{1}{2} \xi_m^T \xi_m + c \quad (1)$$

$$\text{LSE}(\beta, e) = \beta^{-1} \log \left(\sum_{j=1}^{N_i} \exp(\beta e_{ij}) \right) \quad (2)$$

where LSE represents the log-sum-exp function, and β and c are constants. The vector e denotes $\xi_m^T S$, and N_i indicates the number of samples in the i th class. Since the second term $\xi_m^T \xi_m$ represents the regularization of ξ_m , its purpose is to prevent the input pattern from becoming too large during the network update process. The third term c is a constant, and therefore, both terms are unnecessary when evaluating the correlation between the stored patterns and the input pattern. Therefore, the critical component of the energy function is the LSE.

2) *Modality Energy Score*: To estimate modality confidence, we use the Hopfield Energy as a means of quantifying the amount of information in each modality. It evaluates the effectiveness of the current modality based on its similarity to stored modality-specific memory patterns. By simplifying the LSE using a Taylor series, the following expression is obtained:

$$\begin{aligned} \text{LSE}(\beta, e_i) &\approx \frac{1}{\beta} \log \left(\sum_{j=1}^{N_i} (1 + \beta e_{ij}) \right) \\ &= \frac{1}{\beta} \log N_i + \frac{1}{\beta} \log \left(1 + \beta \xi_m^T \frac{\sum_{j=1}^{N_i} s_{ij}}{N_i} \right). \end{aligned} \quad (3)$$

Since all features of the same modality in the same class of samples share the same storage pattern size N_i , the first term can be treated as a constant. By simplifying (3), the energy function is, thus, defined as

$$E(\xi_m) = \frac{1}{\beta} \log (1 + \beta \xi_m^T \bar{s}_i) \quad (4)$$

$$\bar{S}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} s_{ij}. \quad (5)$$

As shown in (4), the key term for confidence computation is $\bar{S}_i$. During the training phase, these modality representations Z_m are treated as stored patterns to compute the modality center using (5). The similarity between the input pattern ξ_m and the modality center is used to quantify the deviation of modality information. $E(\xi_m)$ serves as the confidence weight W_m for the current modality quality. This process aligns the modality centers toward denser regions of the distribution, enabling more accurate estimation of the information content for each input modality. As a result, useful information is better preserved while redundant signals are effectively suppressed.

3) *Maximum Class Probability*: For multimodal data, the amount of information from different modalities is constantly changing. Therefore, it is essential to capture the reliable components of modality information during the classification process. As shown in Fig. 2(a), to extract the reliable components of each modality, we combine the modality confidence W_m obtained from (4) with the modality representation layer to obtain the reliable representation of the modality as follows:

$$Z_m^{\text{conf}} = W_m \cdot Z_m. \quad (6)$$

To obtain more accurate unimodal representations, we design a classifier for each modality, $f_m : x_n^m \rightarrow y_n$. The probability distribution predicted by each modality is obtained through the softmax function, resulting in $p^m(y|X^m) = [p_1^m, \dots, p_K^m]$, where p_k^m represents the predicted probability of class k for the m th modality. The cross-entropy loss function is used to compute the classification error for unimodal data, ensuring that the predicted probability distribution gets closer to the true probability distribution

$$\mathcal{L}_{\text{mcls}} = - \sum_{m=1}^M \sum_{k=1}^K y_k \log p_k^m. \quad (7)$$

During this process, the encoder network E_m generates more discriminative latent representations for each modality in the fusion phase.

B. Cyclic Progressive Fusion for Multimodality

Consistency and complementarity are central to capturing the overall semantic information in multimodal learning. However, enforcing consistency often comes at the cost of overlooking modality-specific information, leading to potential information loss. To address this, we perform gradual fusion of modality features in the representation space, enabling fine-grained integration while preserving distinctive modality cues. As shown in Fig. 2(b), we design a cyclic progressive fusion mechanism to gradually align and integrate information across modality pairs, while adversarial learning is introduced to enforce distribution-level alignment and promote the learning of modality-invariant yet discriminative representations.

Algorithm 1 TCPFMC

Require: Multimodal input samples $\{X_m\}_{m=1}^M$, labels Y
Ensure: Predicted labels $\hat{Y}$

- 1: Initialize modality memory centers $\{\bar{S}_m\}_{m=1}^M = \left\{ \frac{1}{N} \sum_{j=1}^N Z_{mj} \right\}_{m=1}^M$
- 2: **for** each training iteration **do**
- 3: **for** each modality m **do**
- 4: Extract latent representation ξ_m from Z_m
- 5: Compute energy score: $E(\xi_m) = \frac{1}{\beta} \log(1 + \beta \xi_m^T \bar{S}_m)$
- 6: Use $E(\xi_m)$ as confidence weight W_m
- 7: Update memory center $\bar{S}_m$
- 8: **end for**
- 9: **for** each modality pair (i, j) **do**
- 10: Generate $G_{m_i \rightarrow m_j}(Z_{m_i}^{\text{conf}}), G_{m_j \rightarrow m_i}(Z_{m_j}^{\text{conf}})$ compute adversarial loss with D_{m_j}, D_{m_i}
- 11: Update G and D via Eq. 8 and Eq. 9
- 12: **end for**
- 13: Fuse refined features
- 14: Predict label $\hat{Y}$ with final classifier
- 15: **end for**
- 16: **return** $\hat{Y}$

1) *Adversarial Modality Generation*: To preserve modality-specific information, we design a progressive distribution alignment strategy. GANs are effective at aligning distributions through adversarial learning, capturing semantic differences across modalities. We adopt a GAN-based structure to fuse modalities, improving consistency while retaining discriminative features. We take two modalities as an example and design corresponding mapping functions for each modality, denoted as G_{m_i} and G_{m_j} . Specifically, we sequentially select modality pairs from the multimodal representation, with one modality serving as the source modality and the other as the target modality. The purpose of the mapping functions is to map the features of the source modality to the target modality features, i.e., $G_{m_j} : Z_{m_i}^{\text{conf}} \rightarrow Z_{m_j}^{\text{conf}}$ and $G_{m_i} : Z_{m_j}^{\text{conf}} \rightarrow Z_{m_i}^{\text{conf}}$. To ensure that the mapping functions yield a comprehensive representation of the fused target modality information, we design two corresponding discriminators, D_{m_i} and D_{m_j} . Both discriminators adopt a modality-specific design to avoid potential biases caused by parameter sharing across modalities. Specifically, each modality is equipped with an independent multilayer perceptron (MLP)-based discriminator. These discriminators are responsible for distinguishing between $Z_{m_i}^{\text{conf}}$ and $G_{m_i}(Z_{m_j}^{\text{conf}})$, and between $Z_{m_j}^{\text{conf}}$ and $G_{m_j}(Z_{m_i}^{\text{conf}})$, respectively. The objective function is expressed as

$$\begin{aligned} \mathcal{L}_g = & \mathbf{E}_{y \sim p(Z_{m_j}^{\text{conf}})} [\log D_{m_j}(y)] \\ & + \mathbf{E}_{x \sim p(Z_{m_i}^{\text{conf}})} [\log (1 - D_{m_j}(G_{m_j}(x)))] \\ & + \mathbf{E}_{y \sim p(Z_{m_i}^{\text{conf}})} [\log D_{m_i}(y)] \\ & + \mathbf{E}_{x \sim p(Z_{m_j}^{\text{conf}})} [\log (1 - D_{m_i}(G_{m_i}(x)))] . \end{aligned} \quad (8)$$

In the ideal state of the objective function, we provide the discriminator with prior knowledge. The discriminator D_{m_i} should classify the real Z_{m_i} as 1, while classifying $G_{m_j}(Z_{m_i}^{\text{conf}})$,

which is mapped by G_{m_j} , as 0. Similarly, the discriminator D_{m_j} performs the same task for Z_{m_j} and $G_{m_i}(Z_{m_j}^{\text{conf}})$.

2) *Progressive Cycle-Consistency Loss*: To reduce the information loss of modalities during the fusion process by leveraging fine-grained modality correspondences, we design a progressive cycle consistency loss to enforce the consistency between the mapping functions G_{m_i} and G_{m_j} . Specifically, the overall process can be expressed as the forward cycle consistency $Z_{m_i}^{\text{conf}} \rightarrow G_{m_j}(Z_{m_i}^{\text{conf}}) \rightarrow G_{m_i}(G_{m_j}(Z_{m_i}^{\text{conf}})) \rightarrow Z_{m_i}^{\text{conf}}$, and the backward cycle consistency $Z_{m_j}^{\text{conf}} \rightarrow G_{m_i}(Z_{m_j}^{\text{conf}}) \rightarrow G_{m_j}(G_{m_i}(Z_{m_j}^{\text{conf}})) \rightarrow Z_{m_j}^{\text{conf}}$, ensuring that modality information is preserved with minimal loss during fusion. Thus, the objective function can be defined as

$$\mathcal{L}_{\text{cyc}} = \mathbf{E}_{x \sim p(Z_{m_i}^{\text{conf}})} [\|G_{m_i}(G_{m_j}(x)) - x\|_1] + \mathbf{E}_{y \sim p(Z_{m_j}^{\text{conf}})} [\|G_{m_j}(G_{m_i}(y)) - y\|_1]. \quad (9)$$

Through the consistency constraint, the representations obtained from the mapping function G will encompass the characteristics of multiple modalities. This ensures that the fusion among modalities retains useful information while filtering out redundant information. We combine the target modality features generated by G_{m_i} and G_{m_j} , namely $G_{m_j}(Z_{m_i}^{\text{conf}})$ and $G_{m_i}(Z_{m_j}^{\text{conf}})$, to form the final representation Z^{final} . The classifier $f(Z^{\text{final}})$ then generates the predicted probability distribution p_k . Cycle consistency loss effectively prevents mode collapse [64], thereby ensuring the stability of the entire progressive process. The classification objective function is expressed as

$$\mathcal{L}_{\text{cls}} = - \sum_{k=1}^K y_k \log p_k + \mathcal{L}_{\text{mcls}}. \quad (10)$$

In summary, our overall objective function is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_g + \lambda_2 \mathcal{L}_{\text{cyc}} \quad (11)$$

where λ_1 and λ_2 are hyperparameters used to balance the loss function. The overall framework of the proposed method adopts a joint optimization approach to update the parameters at each stage, ensuring the reliability of modality confidence while maintaining the effectiveness of the progressive fusion process.

IV. EXPERIMENTS

This section presents extensive experiments on six real-world multimodal datasets to empirically demonstrate and validate the method's effectiveness.

A. Datasets

We evaluate the performance of TCPFMC on six multimodal datasets: BRCA, ROSMAP, LGG, KIPAN, MVSA, and FOOD101. The BRCA is a five-class classification dataset with 875 samples, commonly used to classify the PAM50 subtype of breast invasive carcinoma. The ROSMAP is a two-class classification dataset with 351 samples, commonly used for diagnosing Alzheimer's disease. The LGG is a two-class classification dataset with 510 samples, commonly used to

TABLE II
DETAILS OF THE SIX MULTIMODAL DATASETS

Datasets	#Train	#Validation	#Test	#Total	#Classes
BRCA	612	-	263	875	5
ROSMAP	245	-	106	351	2
LGG	357	-	153	510	2
KIPAN	460	-	198	658	3
MVSA	1555	518	519	2592	3
FOOD101	62971	5000	22715	90686	101

classify grades in glioma. The KIPAN is a three-class classification dataset with 658 samples, commonly used to classify types of kidney cancer. These four datasets contain three different modalities: DNA methylation (DNA), mRNA expression (mRNA), and miRNA expression (miRNA). Notably, BRCA, ROSMAP, LGG, and KIPAN are widely adopted multiomics benchmarks in multimodal biomedical learning and have been used by mainstream trustworthy fusion methods, enabling fair and reproducible comparisons with prior work. While ROSMAP can be accessed through the AMP-AD Knowledge Portal,¹ KIPAN, LGG, and BRCA are available through The Cancer Genome Atlas (TCGA) program.² The MVSA [65] is a three-class sentiment analysis benchmark that comprises image-text pairs with human annotations, collected from various social media platforms. The FOOD101 [66] is a large-scale dataset comprising 101 categories, containing potentially noisy images collected from Google Image Search and paired with corresponding textual descriptions.

Table II provides a detailed breakdown of the six multimodal datasets. Throughout our experimental process, we strictly adhered to the same data splits as the baseline methods to ensure a fair comparison. Specifically, the data partitioning for the BRCA, ROSMAP, LGG, and KIPAN datasets was consistent with that used in MOGONET [15]; meanwhile, the MVSA and FOOD101 datasets adopted the same splitting methodology as QMF [47].

B. Experimental Settings

We mainly elaborate on the challenges of the datasets and the corresponding experimental design from the following aspects.

1) *Dataset Challenges and Design Motivation*: The datasets used in our experiments exhibit a variety of real-world multimodal challenges.

- 1) To address the large differences in distribution and informativeness across modalities, as seen in BRCA, MVSA, and FOOD101, we design an energy-based confidence mechanism that adaptively adjusts each modality's weight during fusion using memory-guided estimates.
- 2) Some modalities may contain noise or redundant information. For example, methylation signals in ROSMAP or low-saliency visual/textual content in FOOD101. To address this, we introduce adversarial learning

¹<https://adknowledgeportal.synapse.org/>

²<https://gdac.broadinstitute.org/>

TABLE III
COMPARISON OF PERFORMANCE WITH STATE-OF-THE-ART METHODS ON DATASETS BRCA AND KIPAN

Method	Fusion strategy	BRCA			KIPAN		
		ACC	WeightedF1	MacroF1	ACC	WeightedF1	MacroF1
KNN	early	72.1±1.4	71.2±2.2	68.2±2.1	96.6±1.4	96.6±1.4	96.2±1.4
SVM	early	72.3±1.3	71.3±1.4	65.0±1.7	99.3±0.4	99.3±0.4	99.3±0.4
LR	early	73.4±1.4	70.6±1.4	64.9±1.9	97.4±0.3	97.4±0.3	97.3±0.3
RF	early	75.1±1.1	73.8±1.2	65.9±1.6	98.2±0.5	98.2±0.5	97.4±1.1
NN	early	75.4±2.8	74.0±3.3	66.3±3.3	99.1±0.5	99.1±0.5	99.1±0.5
GRidge	intermediate	75.1±1.1	70.2±1.9	68.7±2.1	99.5±0.5	99.5±0.5	99.5±0.5
BPLSDA	intermediate	66.2±1.5	56.2±1.6	38.7±1.7	93.3±1.3	93.3±1.3	91.9±1.8
BSPLSDA	intermediate	64.7±1.3	53.9±1.2	36.1±2.1	91.9±1.2	91.8±1.2	90.2±1.1
CF	intermediate	80.7±1.7	81.1±1.5	79.1±1.1	99.2±0.5	99.2±0.5	98.9±0.5
GMU	intermediate	80.6±2.4	78.2±3.3	72.9±2.9	97.6±1.6	97.6±1.8	96.1±1.7
Unit	intermediate	82.5±2.1	82.4±2.5	78.3±2.9	98.2±0.5	98.2±0.5	98.0±0.4
MOGONET	decision	81.6±1.6	81.7±1.5	76.2±1.6	99.9±0.2	99.9±0.2	99.9±0.2
MoTIF	decision	84.6±2.0	84.0±2.3	74.7±0.5	99.6±0.3	99.6±0.3	99.4±0.4
MLCLNet	decision	86.4±1.6	87.8±1.7	82.6±1.8	99.9±0.7	99.9±0.2	99.9±0.2
TMC	dynamical	85.1±1.3	85.2±1.2	81.0±1.4	99.7±0.3	99.7±0.3	99.5±0.4
QMF	dynamical	79.2±0.8	78.5±0.8	71.7±1.2	100.0±0.0	100.0±0.0	100.0±0.0
MMDynamics	dynamical	87.7±0.3	88.0±0.5	84.5±0.5	99.9±0.2	99.9±0.2	99.9±0.3
DMIB	dynamical	86.2±0.4	86.4±0.4	82.1±0.7	99.9±0.2	99.9±0.2	99.8±0.4
ACMVL	dynamical	79.8±0.7	79.0±0.8	73.6±1.0	99.8±0.3	99.8±0.3	99.6±0.5
Ours	dynamical	88.3±0.5	88.8±0.5	85.8±0.5	100.0±0.0	100.0±0.0	100.0±0.0

with cycle-consistency regularization. This helps reduce cross-modal noise.

- 3) In image–text datasets, such as MVSA and FOOD101, inconsistent cross-modal semantics across samples present a major challenge. To address this, we propose a cyclic progressive fusion strategy that dynamically integrates modalities, enhancing overall representation while preserving modality-specific information.

2) *Implementation Details:* Experiments were conducted on a Linux (Ubuntu 18.04.6) system with an Nvidia A40 GPU and an Intel³ Xeon³ Gold 5320 CPU @ 2.20 GHz. To assess the performance of each method, we used various evaluation metrics for both binary and multiclass datasets. For multiclass datasets, we used accuracy (ACC), weighted average $F1$ score (Weighted- $F1$), and macro-averaged $F1$ score (Macro- $F1$). For binary classification, metrics like accuracy (ACC), $F1$ score ($F1$), and area under the receiver operating characteristic curve (AUC) were applied.

For the omics datasets (e.g., BRCA and LGG), each modality corresponds to a specific omics data type, including mRNA, miRNA, and DNA methylation. We employ separate modality-specific MLP encoders to project each modality into a unified latent space with the same dimensionality. This design ensures modality alignment and compatibility for subsequent fusion. For the MVSA and FOOD101 datasets, we follow the standard feature extraction protocols used in prior works [47] to ensure fair comparison and reproducibility. Specifically, for the image modality, we adopt the widely used ResNet [67] pretrained on ImageNet and extract 2048-dimensional visual features from the penultimate global average pooling layer. For the text modality, we use the widely used BERT-base (uncased) model [68] to tokenize the input and extract a 768-dimensional textual

representation. These fixed features are then fed into our fusion framework without additional fine-tuning.

C. Baselines

To evaluate the performance of TCPFMC, we compared it with 19 other methods, comprising five single-modal approaches and 14 multimodal classification techniques.

The traditional early fusion baselines concatenate features from different modalities as input, including support vector machines (SVM) [25], K -nearest neighbors (KNN) [26], random forests (RF) [27], $L1$ -regularized linear regression (LR), and fully connected NNs. In addition, we compared our method with 14 state-of-the-art multimodal classification techniques.

1) *Methods Based on Intermediate Fusion:* These methods focus on fusing information at the feature or latent representation level, combining modalities before making the final prediction.

- 1) *GRidge* [29]: Group-regularized (logistic) ridge regression, which leverages the structure of multimodal data by applying group-specific regularization.
- 2) *BPLSDA* [30]: Block partial least squares discriminant analysis, which analyzes multimodal data in the latent space using discriminant analysis.
- 3) *BSPLSDA* [30]: Block sparse partial least squares discriminant analysis, which extends BPLSDA by selecting the most relevant features through sparse constraints.
- 4) *CF* [17], [18]: Combining final multimodal representations, which combines modalities by concatenating representations from the final stage.
- 5) *GMU* [31]: Gated multimodal units for information fusion, which produce an intermediate cross-modal

³Registered trademark.

TABLE IV
COMPARISON OF PERFORMANCE WITH STATE-OF-THE-ART METHODS ON DATASETS LGG AND ROSMAP

Method	Fusion strategy	LGG			ROSMAP		
		ACC	F1	AUC	ACC	F1	AUC
KNN	early	73.2±2.8	73.8±3.1	79.6±2.8	66.2±2.7	67.6±3.1	70.3±2.4
SVM	early	74.9±3.5	75.1±3.7	76.2±3.5	77.0±2.2	77.8±2.2	77.0±2.3
LR	early	76.1±2.1	76.7±2.5	82.3±2.5	71.2±3.1	73.0±3.1	77.0±3.5
RF	early	73.9±1.3	74.4±1.4	81.6±1.7	73.4±2.8	73.8±2.9	82.1±2.7
NN	early	73.4±1.8	74.8±1.9	81.1±1.8	76.2±2.1	76.9±1.8	82.7±2.1
GRidge	intermediate	75.2±2.7	76.2±3.4	83.5±3.1	77.2±2.1	77.7±2.2	83.8±2.3
BPLSDA	intermediate	75.5±2.5	74.6±2.3	83.1±2.2	73.2±2.3	75.1±1.7	82.9±2.1
BSPLSDA	intermediate	69.9±2.2	68.4±2.3	76.1±2.2	75.8±2.5	75.9±2.2	83.2±2.7
CF	intermediate	81.2±1.3	82.7±1.7	88.5±1.5	77.7±1.3	77.9±1.5	87.9±1.5
GMU	intermediate	81.3±1.9	81.6±1.7	88.5±1.7	78.4±2.5	78.8±2.3	87.1±2.5
Unit	intermediate	78.7±1.7	78.6±1.8	82.1±1.8	81.5±2.2	81.4±2.2	86.3±2.1
MOGONET	decision	80.7±1.3	80.8±1.7	83.7±1.6	82.8±2.2	82.8±2.3	87.7±2.2
MoTIF	decision	82.2±0.7	81.4±0.8	<u>89.2±0.3</u>	79.3±0.6	78.7±1.7	85.4±0.4
MLCLNet	decision	83.5±1.4	84.0±1.3	88.6±1.2	84.4±1.5	85.2±1.5	89.3±1.1
TMC	dynamical	81.1±1.7	81.9±1.4	87.6±1.4	81.1±1.4	81.8±1.6	88.1±1.6
QMF	dynamical	80.4±1.6	80.3±1.6	87.1±1.4	78.3±1.3	78.3±1.3	85.2±1.2
MMDynamics	dynamical	83.3±1.0	83.7±0.4	88.5±0.4	84.2±1.3	84.6±0.7	91.2±0.7
DMIB	dynamical	83.3±0.8	83.8±0.8	89.1±0.1	<u>84.8±1.4</u>	<u>85.9±1.4</u>	<u>91.6±0.7</u>
ACMVL	dynamical	82.9±1.3	82.9±1.3	86.2±1.0	83.2±1.7	83.2±1.7	89.6±1.0
Ours	dynamical	85.1±0.6	85.5±0.7	89.4±0.1	86.4±1.2	86.3±0.9	92.6±0.9

representation by combining data from different modalities.

- 6) *Unit* [39]: The unified transformer, built on the transformer architecture, effectively shares parameters to maintain high performance.

2) *Methods Based on Decision-Level Fusion*: These methods make independent predictions for each modality and combine them at the decision level, often based on confidence or weighting schemes.

- 1) *MOGONET* [15]: Multiomics graph convolutional networks, which construct a separate GCN for each modality to preserve data structure and capture relationships between modalities through a correlation discovery model.
- 2) *MoTIF* [20]: Multiomics trustworthy integration framework, which enhances the robustness of multimodal learning models by incorporating dynamic feature and modality selection modules.
- 3) *MLCLNet* [16]: Multilevel confidence learning for trustworthy multimodal classification, which leverages a feature confidence mechanism to suppress redundant features.

3) *Methods Based on Dynamic Fusion*: These methods adaptively assess the importance of each modality or feature for each sample, performing fusion in a dynamic manner.

- 1) *TMC* [46]: Trusted multiview classification, which combines decisions relying on the confidence levels of different modalities.
- 2) *QMF* [47]: Quality-aware multimodal fusion (QMF) framework, which enhances robustness by leveraging uncertainty estimation for more effective multimodal fusion.
- 3) *MMDynamics* [22]: Trustworthy multimodal classification through dynamic fusion, which adaptively assesses

the informativeness at both the feature and modality levels for each sample.

- 4) *DMIB* [28]: A dynamic multimodal information bottleneck framework, which improves performance by filtering out noise and preserving task-relevant features in clinical settings.
- 5) *ACMVL* [50]: Adaptive confidence multiview learning method, which enhances multimedia retrieval by fusing multiview confidence and removing noise.

D. Experimental Results

1) *Performance Evaluation Across Multimodal Benchmarks*: TCPFMC achieves the best overall performance on four real-world biomedical multimodal datasets—BRCA, ROSMAP, LGG, and KIPAN. As shown in Table III, TCPFMC demonstrates a significant advantage over unimodal baselines. For instance, on the multiclass classification task of the BRCA dataset, compared to the best unimodal method, TCPFMC achieves absolute improvements of 12.9%, 14.8%, and 17.6% in ACC, Weighted-F1, and Macro-F1, respectively. These results highlight the strength of modality fusion over relying on any single modality. However, even compared with state-of-the-art multimodal fusion approaches, TCPFMC consistently yields the highest performance, showcasing its robustness and superior integration capability across diverse and heterogeneous modalities. In the binary classification tasks on LGG and ROSMAP (see Table IV), TCPFMC maintains its leading performance across all evaluation metrics, further validating its adaptability to different classification settings. To evaluate generalization beyond biomedical domains, we also compare TCPFMC with baselines on two general multimodal benchmarks: MVSA and FOOD101. As illustrated in Table V, TCPFMC exhibits a clear advantage on MVSA, and performs competitively on FOOD101, achieving strong results

TABLE V
COMPARISON OF PERFORMANCE WITH STATE-OF-THE-ART METHODS ON DATASETS MVSA AND FOOD101

Method	Fusion strategy	MVSA			FOOD101		
		ACC	WeightedF1	MacroF1	ACC	WeightedF1	MacroF1
KNN	early	61.2±0.1	58.8±1.5	58.1±3.1	61.4±2.3	61.5±2.5	58.5±2.3
SVM	early	61.1±0.8	54.6±0.2	52.7±0.3	60.9±2.6	61.1±2.3	58.1±2.4
LR	early	62.9±4.3	62.9±4.3	58.2±3.8	61.8±2.5	61.6±2.3	59.7±2.4
RF	early	61.7±0.4	59.4±1.4	58.6±1.9	63.9±2.6	63.8±2.6	60.6±2.3
NN	early	66.3±0.2	66.4±0.2	62.8±0.3	65.3±2.7	65.7±2.3	61.0±2.6
GRidge	intermediate	68.3±1.4	68.1±1.6	65.3±1.2	76.9±1.9	77.1±1.9	76.8±1.6
BPLSDA	intermediate	67.3±1.5	67.4±1.4	64.7±1.5	76.7±1.7	76.5±1.7	74.2±1.2
BSPLSDA	intermediate	66.1±1.2	66.3±1.3	62.8±1.9	75.8±2.3	76.4±2.1	73.1±2.2
CF	intermediate	70.3±1.2	70.5±1.4	66.4±1.3	82.4±0.7	82.4±0.8	82.4±0.6
GMU	intermediate	73.7±2.9	73.4±3.1	69.2±3.2	81.7±2.1	81.8±2.0	81.7±2.1
Unit	intermediate	78.5±1.8	78.2±1.8	76.1±1.8	87.2±0.2	87.5±0.2	87.4±0.2
MOGONET	decision	64.7±1.9	65.7±5.6	64.1±5.8	84.3±2.6	84.2±2.7	84.1±2.7
MoTIF	decision	66.9±0.9	63.6±3.0	61.1±6.9	84.7±2.1	84.6±2.0	84.1±2.1
MLCLNet	decision	72.8±1.8	72.1±1.6	70.1±1.8	84.8±2.4	84.7±2.6	84.9±2.8
TMC	dynamical	74.4±1.0	73.9±1.2	70.4±1.6	89.8±0.8	89.8±0.8	89.9±0.8
QMF	dynamical	78.2±1.1	77.7±1.2	74.0±1.2	92.8±0.4	92.7±0.4	92.6±0.4
MMDynamics	dynamical	76.4±1.7	75.8±1.5	72.3±1.7	87.1±2.5	87.2±2.7	87.1±2.5
DMIB	dynamical	70.4±6.8	69.4±7.0	65.6±7.3	86.1±1.6	87.2±1.4	87.1±1.4
ACMVL	dynamical	78.0±1.9	77.4±2.1	74.6±3.1	89.9±1.1	89.9±1.1	90.0±1.2
Ours	dynamical	81.7±0.5	81.7±0.7	78.9±1.2	92.7±0.2	92.8±0.2	92.7±0.2

TABLE VI
P-VALUES OF ACCURACY COMPARISONS ACROSS DIFFERENT
FUSION ALGORITHMS ON FOUR DATASETS

Fusion strategy	BRCA	KIPAN	LGG	ROSMAP
Early	1.4e ⁻⁴	1.3e ⁻⁴	1.3e ⁻⁴	1.4e ⁻⁴
Intermediate	4.0e ⁻⁵	4.7e ⁻⁵	7.6e ⁻⁵	3.9e ⁻⁵
Decision	0.004	0.029	0.002	0.001
Dynamical	1.7e ⁻⁴	0.225	5.7e ⁻⁴	2.5e ⁻⁴

on Weighted- $F1$ and Macro- $F1$, even though the accuracy is marginally lower than the best baseline. These results confirm that our proposed method not only performs well in complex, high-dimensional medical data but also generalizes effectively to conventional multimodal scenarios.

As Tables III and IV show, the performance of TCPFMC is relatively close to some of the baselines. Therefore, we further conduct a statistical significance analysis based on the Friedman test [69]. The resulting p -values quantify the probability that the observed performance differences among methods occur by chance; the smaller the p -value, the stronger the evidence that the differences are statistically significant. Table VI reports these p -values across the four datasets. For BRCA, LGG, and ROSMAP, all p -values are far below 0.05, indicating that the differences among fusion algorithms are statistically significant on these datasets. In contrast, on KIPAN, the p -value for the dynamical fusion method increases to 0.225 (>0.05), suggesting no statistically significant difference among methods on this dataset. This is expected because almost all algorithms, including ours, achieve near-perfect accuracy on KIPAN, which leads to very small variance and a ceiling effect that makes it difficult for the statistical test to detect differences.

2) *Complexity Analysis*: The computational cost of TCPFMC mainly arises from modality encoders and the

cyclic progressive fusion module. Each modality is processed by an MLP encoder with hidden dimension H , resulting in a complexity of $O(Nd_mH)$ for the m th modality, where N denotes the number of samples, and d_m denotes the feature dimension. Summing over all M modalities gives $O\left(N\sum_{m=1}^M d_mH\right)$. The cyclic progressive fusion further introduces pairwise alignment across modalities, requiring $O(M^2)$ generator-discriminator pairs, each implemented as MLPs, which adds an overhead of $O(M^2H^2)$. Since the number of modalities M is usually small (two or three in our datasets), this overhead is negligible, and the overall complexity can be expressed as $O\left(N\sum_{m=1}^M d_mH + M^2H^2\right)$. Compared with baselines, TCPFMC has the same asymptotic order as other NN-based fusion methods (e.g., MoTIF [20], MMDynamics [22], TMC [46], CF [17], [18], QMF [47], DMIB [28], ACMVL [50], and GMU [31]), and is more efficient than graph-based approaches (e.g., MOGONET [15], MLCLNet [16]) that require $O(N^2D)$ graph construction, or statistical fusion methods (e.g., BPLSDA [30], GRidge [29]) with $O(MND^2)$ or $O\left(N\left(\sum d_m\right)^2\right)$. Transformer-based methods (e.g., Unit [39]) involve $O(M^2H)$ complexity due to self-attention, which scales less favorably with increasing M . In summary, TCPFMC maintains computational efficiency by scaling linearly with data size and feature dimensions while incurring only a minor quadratic cost with respect to the small number of modalities.

3) *Performance of TCPFMC Under Different Modalities Data Types*: To verify the necessity of multimodal integration and the effectiveness of progressive fusion, we perform fusion of different modalities on the four datasets. We selected two combinatorial modalities for experimentation in each dataset, i.e., mRNA + meth, mRNA + miRNA, meth + miRNA, and all combinatorial modalities. The results of the experiments are shown in Fig. 3, where the effect of meth + miRNA

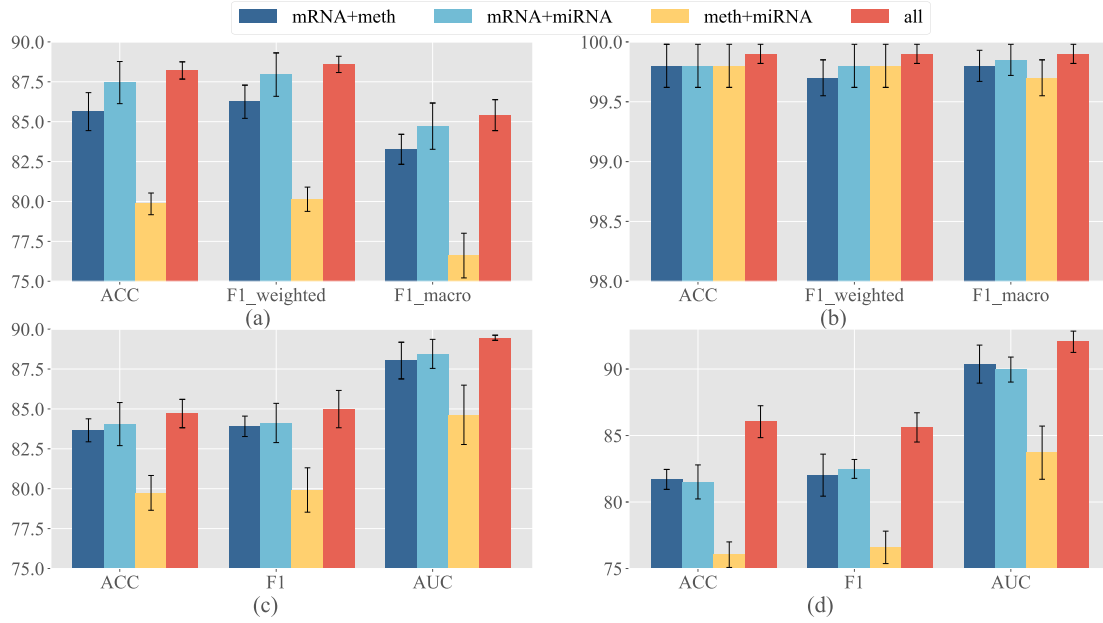


Fig. 3. Classification performance of the proposed method across different modalities. (a) BRCA. (b) KIPAN. (c) LGG. (d) ROSMAP.

TABLE VII
ABLATION STUDY ON CONFIDENCE ESTIMATION AND PROGRESSIVE ADVERSARIAL FUSION

Method	BRCA			KIPAN		
	ACC	WeightedF1	MacroF1	ACC	WeightedF1	MacroF1
OF	81.3±0.6	82.3±0.7	78.7±1.9	99.5±0.3	99.5±0.3	99.4±0.5
CF	86.6±0.7	86.4±0.8	82.2±1.1	99.6±0.2	99.6±0.2	99.4±0.4
MF	87.6±0.5	88.0±0.5	84.7±0.9	99.0±0.2	99.0±0.2	99.0±0.2
TCPFMC	88.3±0.5	88.8±0.5	85.8±0.5	100.0±0.0	100.0±0.0	100.0±0.0
Method	LGG			ROSMAP		
	ACC	F1	AUC	ACC	F1	AUC
OF	81.2±0.9	82.7±0.9	87.8±0.8	79.1±0.7	80.1±0.6	88.5±1.4
CF	83.0±1.2	83.1±0.9	88.1±1.2	82.6±2.6	83.0±1.4	90.1±2.4
MF	83.4±0.7	84.0±0.9	88.6±0.9	84.7±2.0	85.0±1.0	90.2±0.8
TCPFMC	85.1±0.6	85.5±0.7	89.4±0.1	86.4±1.2	86.3±0.9	92.6±0.9
Method	MVSA			FOOD101		
	ACC	WeightedF1	MacroF1	ACC	WeightedF1	MacroF1
OF	71.7±0.8	71.7±0.9	67.8±0.8	83.5±1.1	83.7±1.2	84.3±1.2
CF	78.9±0.4	78.0±0.4	75.4±0.5	91.6±0.8	91.5±0.6	91.6±0.8
MF	77.1±0.6	76.8±0.6	73.9±0.6	84.1±0.9	85.5±0.8	85.4±0.9
TCPFMC	81.7±0.5	81.7±0.7	78.9±1.2	92.7±0.2	92.8±0.2	92.7±0.2

combination is the lowest in all datasets. In most of the datasets, the effect of the mRNA + miRNA combination was suboptimal. This demonstrates that the meth modality contributes less to the multimodal fusion compared to the other two modalities. However, through progressive fusion, we can fully exploit the useful information from the meth modality, resulting in the optimal fusion performance of the three modalities. This further confirms that multimodal progressive fusion can effectively extract useful information from each modality individually. Thus, integrating more modalities enables the extraction of more discriminative semantic information.

E. Ablation Study

1) *Ablation Study on Energy-based Confidence and Progressive Fusion:* To investigate the contribution of each

component in our framework, we conduct a comprehensive ablation study by comparing four model variants: 1) OF, which directly concatenates modality features without energy-based confidence or progressive fusion; 2) CF, which introduces energy-based modality confidence but still uses naive concatenation; 3) MF, which adopts our proposed cyclic progressive fusion strategy but omits energy-based confidence weighting; and 4) TCPFMC, which integrates both components along with adversarial alignment.

As shown in Table VII, both CF and MF outperform the baseline OF across all datasets and metrics, demonstrating that energy-based confidence estimation and progressive fusion are individually beneficial. Specifically, comparing OF and CF, we observe consistent improvements (e.g., +7.2% ACC and +6.3% Weighted-F1 on MVSA), validating that the confidence scores derived from the energy function can effectively

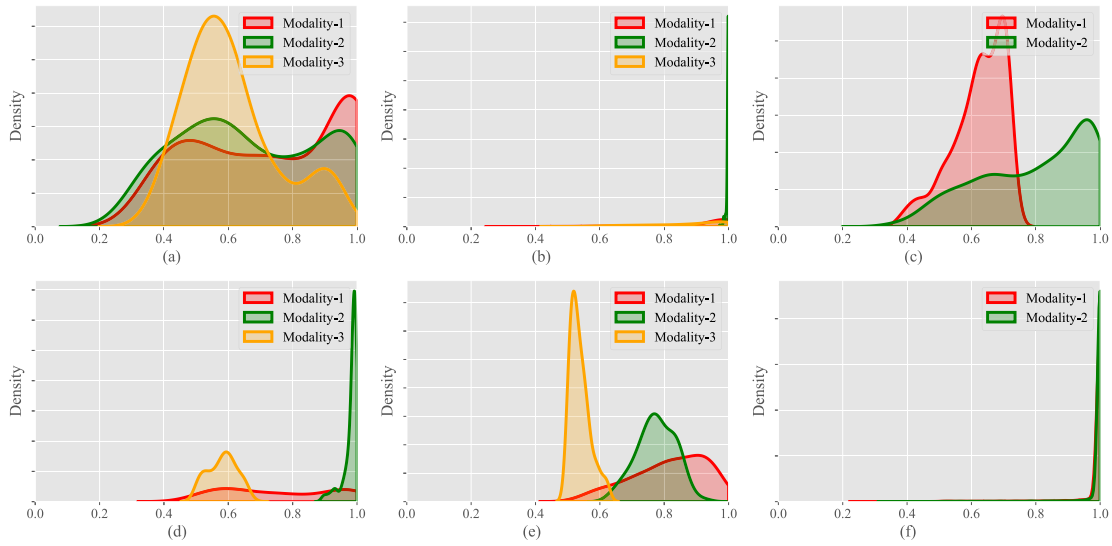


Fig. 4. Modal confidence density analysis using the TCPFMC algorithm. (a) BRCA. (b) KIPAN. (c) MVSA. (d) LGG. (e) ROSMAP. (f) Food101.

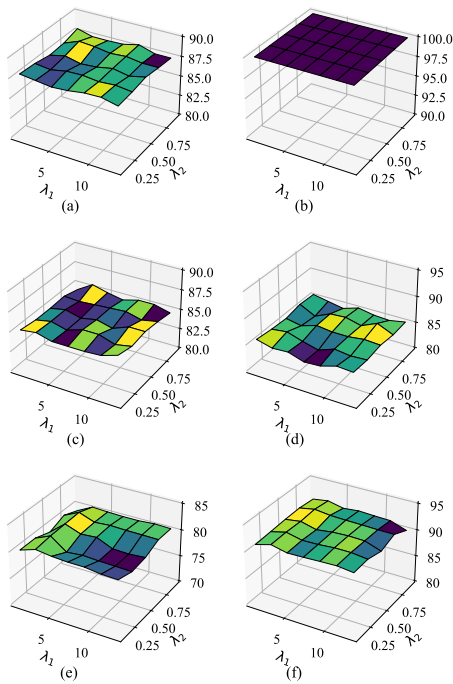


Fig. 5. Parameter sensitivity experiments are conducted on six datasets to investigate the effect of variations in λ_1 and λ_2 on the final classification performance. (a) BRCA. (b) KIPAN. (c) LGG. (d) ROSMAP. (e) MVSA. (f) Food101.

capture the informativeness of each modality and guide the fusion process. Comparing OF and MF, performance also improves significantly (e.g., +6% Macro- F_1 on BRCA and +1.7% AUC on ROSMAP), confirming that progressive, pairwise fusion provides a more structured and robust integration strategy than naive concatenation. Importantly, TCPFMC, which combines both modules, achieves the best performance across all datasets. The results demonstrate that energy-based confidence estimation, progressive fusion, and adversarial alignment work in a mutually complementary manner to

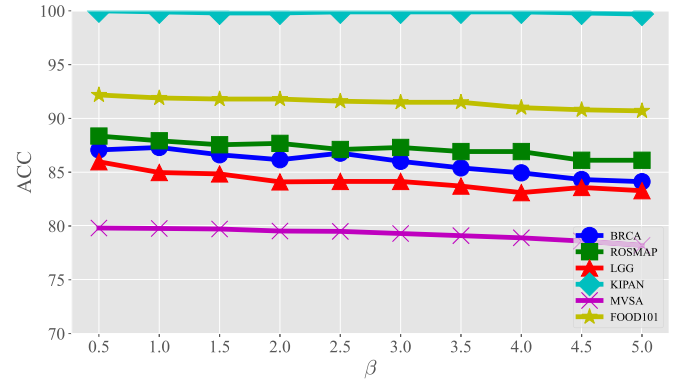


Fig. 6. Parameter sensitivity experiments are conducted on six datasets to investigate the effect of varying β on the final classification performance.

achieve robust, adaptive, and semantically consistent multi-modal learning.

F. Confidence Analysis

To better understand the behavior of the proposed modality energy function, we visualize the density distributions of energy scores for each modality across six datasets, as shown in Fig. 4. In our formulation, a higher energy score indicates greater modality confidence, and a sharper density peak near 1.0 reflects that the modality is consistently reliable across samples. Several key observations can be drawn from the plots. First, in datasets such as BRCA and LGG, different modalities exhibit distinct confidence distributions, indicating varying levels of informativeness and supporting the need for adaptive modality weighting. Second, in the MVSA dataset—a sentiment analysis task—modality 1 (text) shows a strong, high-confidence peak compared to modality 2 (image), suggesting that textual cues are more semantically aligned with the task. Third, in ROSMAP, only modality 3 demonstrates a concentrated high-confidence region, while the others appear more dispersed, implying the presence of noisy or redundant signals. Finally, in KIPAN and Food101, all

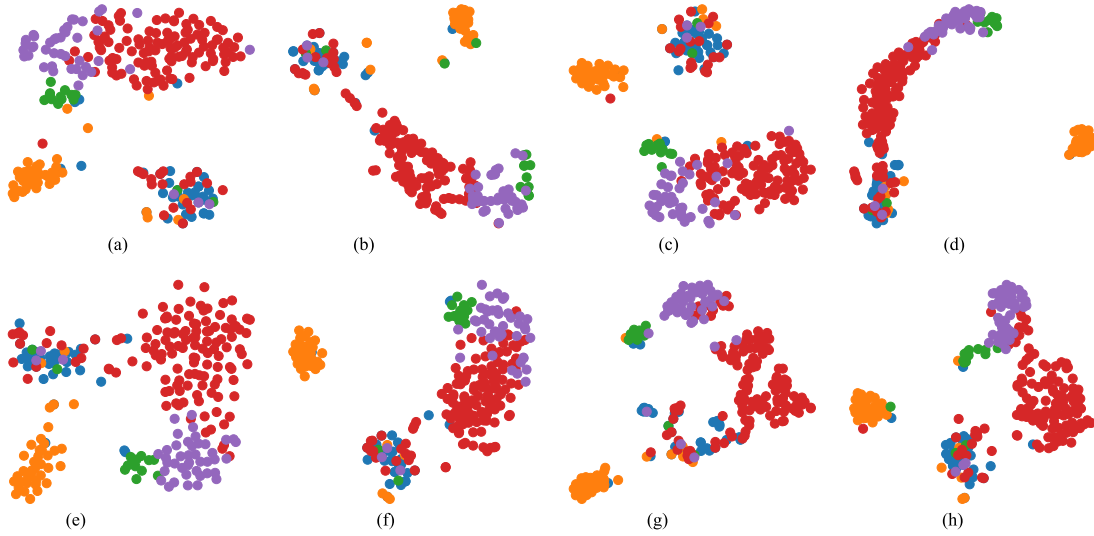


Fig. 7. Evaluation of the quality of latent representations of different methods on the dataset BRCA using t-SNE. (a) CF. (b) MOGONET. (c) GMU. (d) Unit. (e) MMDynamics. (f) DMIB. (g) ACMVL. (h) Ours.

modalities display consistently high confidence, which helps explain the saturated classification performance observed on these datasets. These results confirm that the proposed energy-based scoring mechanism effectively captures modality-level reliability and provides a robust foundation for confidence-aware fusion in diverse multimodal scenarios.

G. Parameter Study

To investigate the sensitivity of TCPFMC to parameters, a parameter sensitivity experiment was conducted, focusing on λ_1 and λ_2 in the loss function. As shown in Fig. 5, the impact of these hyperparameters on model accuracy was evaluated across six datasets. The experimental results indicate that the model performance did not fluctuate significantly with changes in hyperparameters. To further investigate the impact of progressive fusion weighting on overall performance, we fix the value of λ_1 and analyze how varying λ_2 affects the results. As shown in Fig. 5, most datasets achieve peak performance when $\lambda_2 \approx 0.75$. This indicates that appropriately increasing the weighting of later fusion stages helps the model better integrate discriminative information from different modalities, thereby improving overall classification performance. These findings also validate the effectiveness of the progressive fusion strategy in multimodal learning.

To investigate the impact of the parameter β in the modality energy score on multimodal fusion classification, we conducted a series of experiments by varying β . Specifically, β was set to values ranging from 0.5 to 5. As shown in Fig. 6, in most datasets, the model performance remains relatively stable as the parameter β varies. While slight fluctuations may occur on a few datasets due to their specific characteristics, the overall variation is minimal. The performance on all six datasets consistently decreases to varying degrees as β increases. These results indicate that smaller values of β generally lead to better overall performance.

TABLE VIII
QUANTITATIVE EVALUATION OF LATENT FEATURE CLUSTER QUALITY USING STANDARD UNSUPERVISED METRICS

Method	Silhouette ($\uparrow$)	CH ($\uparrow$)	DB ($\downarrow$)
CF	0.22	83.55	1.19
MOGONET	0.25	87.11	1.71
GMU	0.23	92.07	1.11
Unit	0.28	130.18	1.79
MMDynamics	0.28	111.76	0.96
DMIB	0.21	152.68	1.14
ACMVL	0.26	138.30	1.03
Ours	0.32	153.28	0.86

H. Visualization

To qualitatively assess the discriminative power and structure of the learned multimodal representations, we visualize the feature distributions of different methods using t-SNE in Fig. 7. Each point represents a sample in the latent space, color-coded by ground-truth class labels. As shown in the figure, the proposed method (h) achieves significantly better class separation compared to baseline methods. In particular, our approach produces compact intraclass clusters and clear interclass boundaries, indicating strong class-specific consistency and improved modality alignment. By contrast, methods, such as CF (a), GMU (c), and MMDynamics (e) exhibit considerable class overlap, suggesting less effective feature fusion. Although some recent methods, like DMIB (f) and ACMVL (g), show partial clustering improvements, their boundaries remain fuzzy and prone to confusion across neighboring classes. While the visual differences among methods may appear subtle, partly due to the variability of t-SNE projections and their limited ability to capture high-dimensional cluster structures, we therefore quantify the clustering quality using Silhouette score, Calinski–Harabasz index (CH), and Davies–Bouldin Index (DB) (see Table VIII). These quantitative results show that our method consistently

outperforms the baselines, further confirming the effectiveness of our fusion and representation learning strategy.

V. CONCLUSION

In this work, we propose a trustworthy multimodal fusion classification method called TCPFMC. It effectively integrates the modality-specific confidence information to reduce modality bias, while ensuring the stability of the fusion process. In the modality confidence evaluation phase, the proposed concise modality energy score effectively assesses modality confidence and leverages the contextual information of each modality to estimate modal information. In the fusion phase, we design a cyclic progressive fusion approach that gradually integrates modality information, reducing modality-specific information loss during the fusion process. Extensive experiments validate that the proposed method effectively integrates useful information across modalities and show that our approach achieves optimal performance compared to state-of-the-art methods, confirming the promising potential of trustworthy multimodal learning. In future work, we plan to extend TCPFMC to more flexible scenarios involving incomplete or missing modalities.

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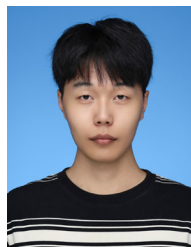
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